Uniwersytet SWPS

Task list no. 5 – In this lab, we will explore the concepts of heap trees, heap sort, and priority queues, and examine how priority queues can be implemented using heaps.

Task 1. Implement the MAX-HEAPIFY algorithm and apply it to a heap in which one node violates the max-heap property.

Task 2. Use MAX-HEAPIFY to build a max-heap.

Task 3. Use the BUILD-MAX-HEAP algorithm to implement the HEAPSORT algorithm.

Note. The lab report should contain your code along with the results.

Note. ALL PROGRAMS SHOULD COMPILE WITHOUT ERROR AND WORK (i.e. DO EXACTLY WHAT IS DESCRIBED IN THE TASK)!!!

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